

CANONICAL LESSON

Climate Signals Before Disease Surges

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C28
Notebook	V7_B_N08
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Use climate signals for disease preparedness only when exposure, lag, surveillance quality, and causal limits are explicit.

Learning outcomes

Why this matters

Climate association alone does not prove a disease surge or justify a specific intervention.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

Rainfall and temperature anomalies inform readiness after epidemiological and geographic corroboration.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Add temperature.
- 2.
- Prevent temporal leakage.
- 3.
- Explain causality limits.

Answer key

- 1.
- Add dated exposures and nonlinear/lagged terms with domain justification.
- 2.
- Fit baselines and choose parameters only on prior data, then evaluate forward.
- 3.
- Shared seasonality, reporting, interventions, ecology, and confounding can create association; causal claims require stronger design and evidence.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.